🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 8 · STANDARD 6.AT.8

Write and Solve Equations Using Multiplication or Division

Lesson 8-3 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

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My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can solve one-step multiplication and division equations using inverse operations.

Language Objective: I can explain my steps using the words inverse operation, multiply, divide, and isolate.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Inverse operation | **Multiply** | **Divide** | **Isolate** | **Coefficient** | **Solution**

👆 *Tap any word to see its meaning and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. An operation that undoes another, like division undoing multiplication, is an ______.

2. To undo dividing by a number, I ______ both sides by that number.

3. To undo multiplying by a number, I ______ both sides by that number.

4. To get the variable alone on one side is to ______ it.

5. The number multiplied by a variable is the ______.

6 The value of the variable that makes an equation true is the

________________.

Write About the Math

How many kits did the detective order? How did inverse operations help you solve this?

¿Cuántos kits pidió la detective? ¿Cómo te ayudaron las operaciones inversas a resolver esto?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Inverse operation** — *Operación inversa*
- ☐ **Multiply** — *Multiplicar*
- ☐ **Divide** — *Dividir*
- ☐ **Isolate** — *Despejar*
- ☐ **Coefficient** — *Coeficiente*

Model: *In 3x = 21 the variable is multiplied, so I undo it with division.* — Students read 3x as 3 groups of x, recognize multiplication, and contrast it with addition, knowing they will divide (not subtract) to solve.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The detective ordered ____ kits because 7k = 63 means k = 63 ÷ ____ = ____.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

____

____

____

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Inverse operation
2. **Notes 2:** Multiply
3. **Notes 3:** Divide
4. **Notes 4:** Isolate
5. **Notes 5:** Coefficient
6. **Notes 6:** Solution
7. **Watch for this mistake:** *A common mistake in Solve Multiplication and Division Equations is applying the SAME operation again instead of the inverse — for example, seeing 6x = 42 and multiplying both sides by 6 to get x = 252, instead of dividing by 6 to get x = 7. Before you submit, ask: did I use the OPPOSITE operation shown in the equation, and did I apply it to both sides?*
8. **Try It 1:** A. x = 18 *— x is divided by 3, so multiply both sides by 3: x = 6 x 3 = 18. Check: 18 / 3 = 6.*
9. **Try It 2:** A. n = 9 *— The 5 is multiplying n, so divide both sides by 5: n = 45 / 5 = 9. Check: 5 x 9 = 45.*